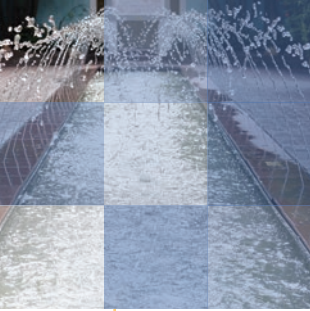




Factors and Rational Roots

Synthetic division can help you factor polynomials.

Factor Theorem

For any polynomial $p(x)$, the binomial $x - a$ is a factor of $p(x)$ if and only if $p(a) = 0$.

The remainder theorem is used to prove the factor theorem.

For the proof, use $p(x)$ for the dividend, $q(x)$ for the quotient, and $r(x)$ for the remainder. Therefore, $p(x) = q(x) \cdot (x - a) + r(x)$.

Because the theorem uses “if and only if,” there are two cases.

Case 1 Assume $p(a) = 0$. Then by the remainder theorem, $r(a) = 0$.

Because the quotient does not have a remainder, $x - a$ is a factor of the dividend.

Case 2 Assume $x - a$ is a factor. Then the remainder $r(a)$ equals 0. By the remainder theorem, $r(a) = p(a)$, and $0 = p(a)$ by substitution.

Using the Factor Theorem

So what does the factor theorem mean? It means that if you evaluate a polynomial for some value of x and the result is zero, then you can use that value to factor the polynomial.

EXAMPLE 1

Determine whether $x - 3$ is a factor of $f(x) = 2x^4 + 2x^3 - 11x^2 + x - 6$.

SOLUTION

Since $x + 3 = x - (-3)$, $a = -3$. So find $f(-3)$.

$$\begin{aligned}f(-3) &= 2(-3)^4 + 2(-3)^3 - 11(-3)^2 + (-3) - 6 \\&= 2 \cdot 81 + 2 \cdot (-27) - 11 \cdot 9 - 3 - 6 \\&= 162 - 54 - 99 - 3 - 6 \\&= 0\end{aligned}$$

Since $f(-3) = 0$, $x + 3$ is a factor of $f(x) = 2x^4 + 2x^3 - 11x^2 + x - 6$. ■

For any polynomial function $p(x)$, a solution of the equation $p(x) = 0$ is called a zero or root of $p(x)$. In Example 1, you could say that -3 is a zero of $f(x)$.

EXAMPLE 2

Factor the polynomial $f(x) = x^3 - 19x - 30$, given that $x = 5$ is a zero.

SOLUTION

Step 1 If $x = 5$ is a zero, then $x - 5$ is a factor. Use synthetic division to divide $x^3 - 19x - 30$ by $x - 5$.

$$\begin{array}{r|rrrr} 5 & 1 & 0 & -19 & -30 \\ & & 5 & 25 & 30 \\ \hline & 1 & 5 & 6 & 0 \end{array}$$

Step 2 Write the polynomial as a product of $x - 5$ and the quotient from Step 1. Then factor the quotient.

$$\begin{aligned}x^3 - 19x - 30 &= (x - 5)(x^2 + 5x + 6) \\&= (x - 5)(x + 3)(x + 2)\end{aligned}$$

So $f(x) = x^3 - 19x - 30 = (x - 5)(x + 3)(x + 2)$. ■

EXAMPLE 3

Find all the roots of $P(x) = x^3 - 19x - 30$.

SOLUTION

By the factor theorem, if $x - a$ is a factor of $P(x)$, then $P(a) = 0$.

Example 2 shows that $x^3 - 19x - 30 = (x - 5)(x + 3)(x + 2)$.

Therefore 5, -3 , and -2 are the roots of $P(x) = x^3 - 19x - 30$. ■

► **Remember** A root is a value that makes a polynomial zero.

Finding Possible Rational Roots

It's easy to find all the roots if you are given a head start, but what if you aren't given any of the roots? How can you find all the roots on your own? The rational root theorem can give you a set of values to try.

Rational Root Theorem

If $\frac{p}{q}$ is in simplest form and is a rational root of the polynomial

$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ with integer coefficients,
then p must be a factor of a_0 and q must be a factor of a_n .

► **Remember** A rational number is a real number that can be written as a quotient of two integers.

EXAMPLE 4

Find the possible rational roots of the polynomial.

A $f(x) = 2x^3 + 16x^2 + 19x + 5$

SOLUTION

If $\frac{p}{q}$ is a rational root of the polynomial $2x^3 + 16x^2 + 19x + 5$, then p is a factor of the constant term, 5, and q is a factor of the leading coefficient, 2. List the factors of 5 and the factors of 2.

The factors of 5 are ± 1 and ± 5 .

The factors of 2 are ± 1 and ± 2 .

The possible roots of $\frac{p}{q}$ are $\pm \frac{1}{1}$, $\pm \frac{1}{2}$, $\pm \frac{5}{1}$, and $\pm \frac{5}{2}$.

Simplifying gives the following possible rational roots of

$$2x^3 + 16x^2 + 19x + 5 = 0:$$

$$\pm 1, \pm \frac{1}{2}, \pm 5, \text{ and } \pm \frac{5}{2}$$

B $g(x) = 3x^3 - 11x^2 + 20x - 9$

SOLUTION

If $\frac{p}{q}$ is a rational root of the polynomial $3x^3 - 11x^2 + 20x - 9$, then p is a factor of the constant term, -9 , and q is a factor of the leading coefficient, 3. List the factors of -9 and the factors of 3.

The factors of -9 are ± 1 , ± 3 , and ± 9 .

The factors of 3 are ± 1 and ± 3 .

The possible roots of $\frac{p}{q}$ are $\pm \frac{1}{1}$, $\pm \frac{1}{3}$, $\pm \frac{3}{1}$, $\pm \frac{3}{3}$, $\pm \frac{9}{1}$, and $\pm \frac{9}{3}$.

Simplifying gives the following possible rational roots of

$$3x^3 - 11x^2 + 20x - 9 = 0:$$

$$\pm 1, \pm \frac{1}{3}, \pm 3, \text{ and } \pm 9 \quad \blacksquare$$

Using the Rational Root Theorem to Factor a Polynomial

EXAMPLE 5

Factor completely.

A $x^3 - x^2 - 10x - 8$

SOLUTION

Identify the possible rational roots: ± 1 , ± 2 , ± 4 , and ± 8 .

Test for possible roots.

$$\begin{array}{r|rrrrr} -1 & 1 & -1 & -10 & -8 & \\ & & -1 & 2 & 8 & \\ \hline & 1 & -2 & -8 & 0 & \end{array}$$

Because -1 is a root, $(x + 1)$ is a factor. The other factor is $x^2 - 2x - 8$, which can be factored: $(x - 4)(x + 2)$.

The factored form is $(x + 1)(x - 4)(x + 2)$.

► **Think About It** You may need to test several roots before finding one that makes the remainder 0. Test easier numbers such as -1 and 1 before testing larger numbers and fractions.

B $x^3 - 3x^2 - 5x + 15$

SOLUTION

Identify the possible rational roots: ± 1 , ± 3 , ± 5 , and ± 15 .

Test for possible roots.

$$\begin{array}{r|rrrrr} & 1 & -3 & -5 & 15 & \\ 1 & 1 & -2 & -7 & 8 & \\ -1 & 1 & -4 & -1 & 16 & \\ 3 & 1 & 0 & -5 & 0 & \end{array}$$

► **Think About It** Organize your work in a table. List the coefficients of the polynomial at the top. Perform the steps mentally and write only the numbers that appear in the bottom row.

Because 3 is the first possible root with a remainder of 0, $(x - 3)$ is a factor. The other factor is $x^2 - 5$. Solve $x^2 - 5 = 0$ to find the other roots.

$$\begin{aligned}x^2 - 5 &= 0 \\x^2 &= 5 \\x &= \pm\sqrt{5}\end{aligned}$$

The factored form is $(x - 3)(x + \sqrt{5})(x - \sqrt{5})$.

C $3x^3 + x^2 - 38x + 24$

SOLUTION

The possible rational roots are $\pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1, \pm\frac{4}{3}, \pm 2, \pm\frac{8}{3}, \pm 3, \pm 4, \pm 6, \pm 8, \pm 12$, and ± 24 .

	3	1	-38	24
1	3	4	-34	-10
-1	3	-2	-36	54
2	3	7	-24	-24
-2	3	-5	-28	80
$\frac{2}{3}$	3	3	-36	0

Testing the possible roots by synthetic division reveals that $\frac{2}{3}$ is a root.

So $(x - \frac{2}{3})$ is a factor. The other factor is $3x^2 + 3x - 36$, which factors into $3(x + 4)(x - 3)$.

The factored form is $3(x - \frac{2}{3})(x + 4)(x - 3)$, which can also be written as $(3x - 2)(x + 4)(x - 3)$. ■